

My Little Library

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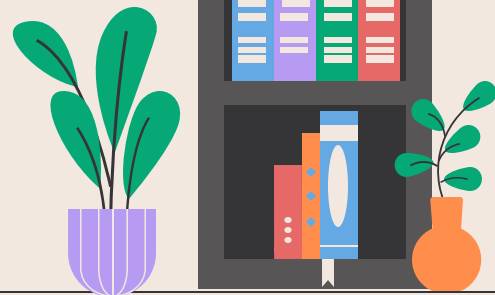




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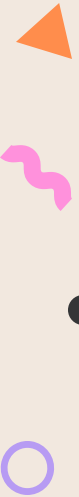
ADD TO LIBRARY

Two pages with different options to store a book

06

GENERATE STORY

A page that generates a story based off reading level





01 | WHAT IS IT?

Originally a project for our **CS322: Data Science in Practice** class, we developed *My Little Library* as an interactive reading and recommendation platform designed to help users track books and receive book suggestions based on reading level.



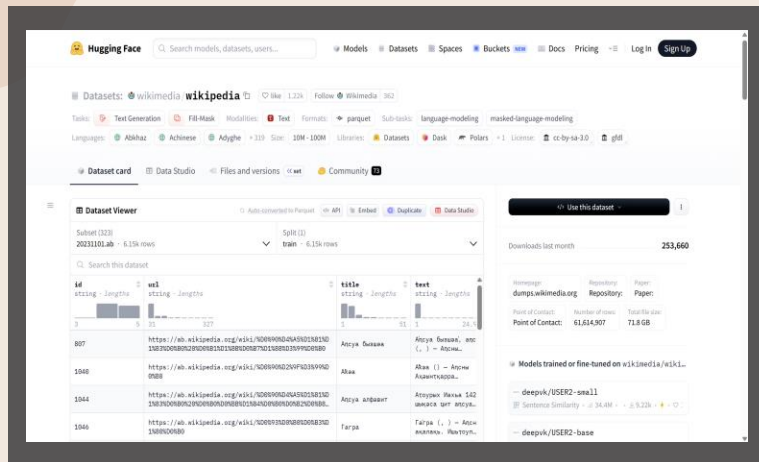
01 | WHAT IS IT?

To support personalized recommendations and story generation, the system utilizes a Retrieval-Augmented Generation (RAG) pipeline powered by the Meta Llama 3.1 8B model and a dataset of nearly 1 million scraped Wikipedia articles.

Coding Languages:

- Python
 - HTML/CSS
 - Javascript
 - Flask
 - SQLite
- 





OUR DATASET

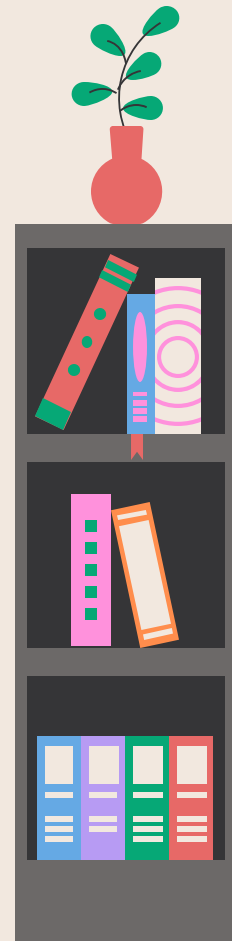
We used HuggingFace's `wikipedia/wikipedia` dataset, allowing us to scrape ~1 million articles.

Link:
<https://huggingface.co/datasets/wikipedia/wikipedia>

02

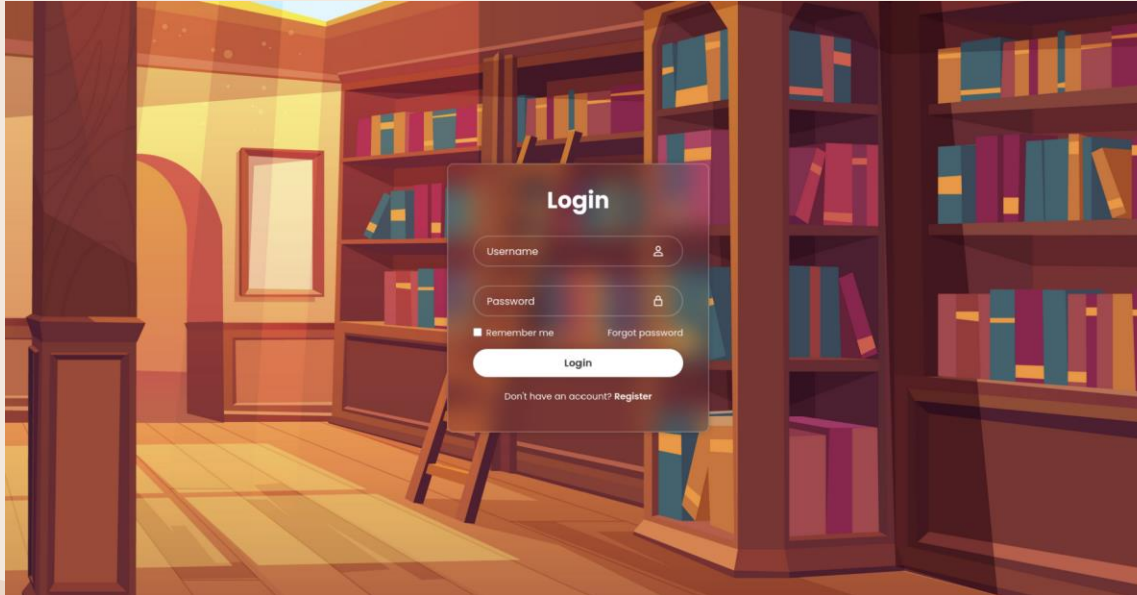
LOGIN PAGE

System's Login & Registration Pages



LOGIN PAGE

When you first open the program, you're greeted by the login page.



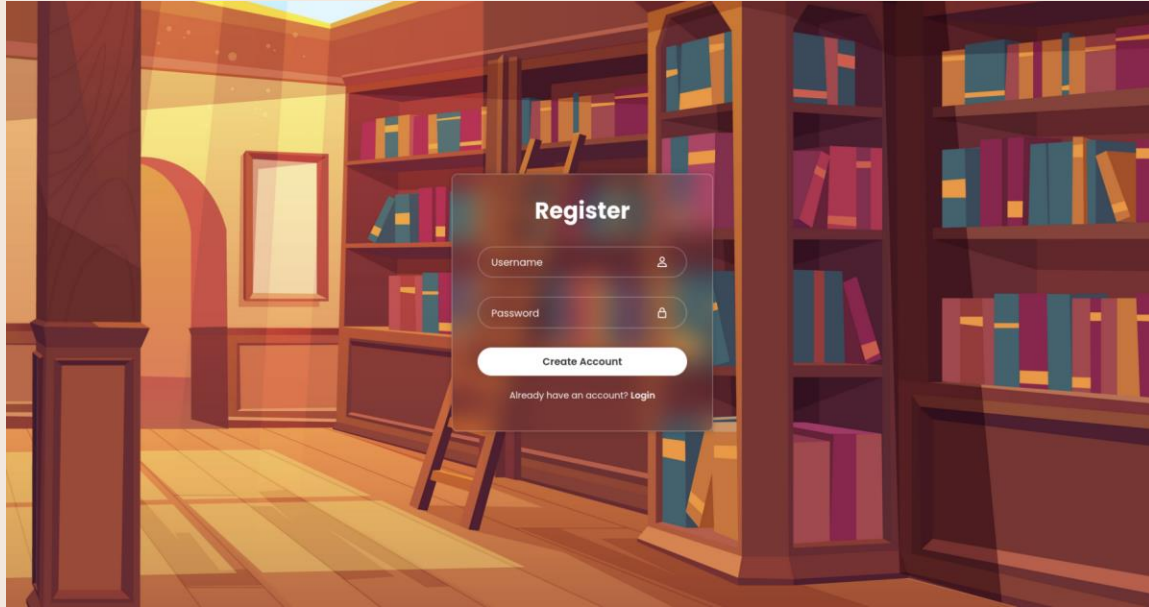
To successfully log in, the user will need to enter their username and password. If they leave one blank, they'll be notified.

If their information is matched with one in the database, then the user will be brought into the main page. If not, they'll receive an error

If the user doesn't have an account, they can click "Register"

REGISTRATION PAGE

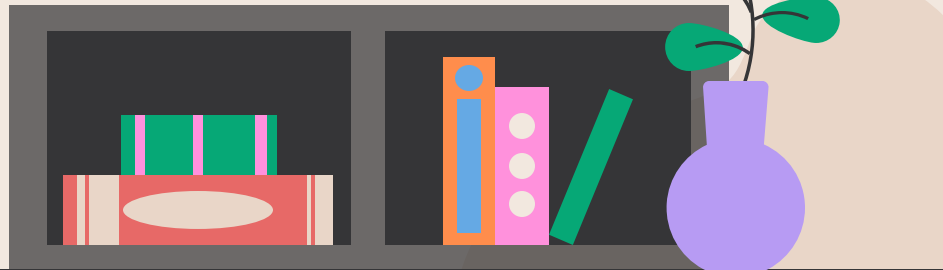
Similar to the login page in appearance, the user can enter a username and password to register their account with.



Their account information is then stored into the database.

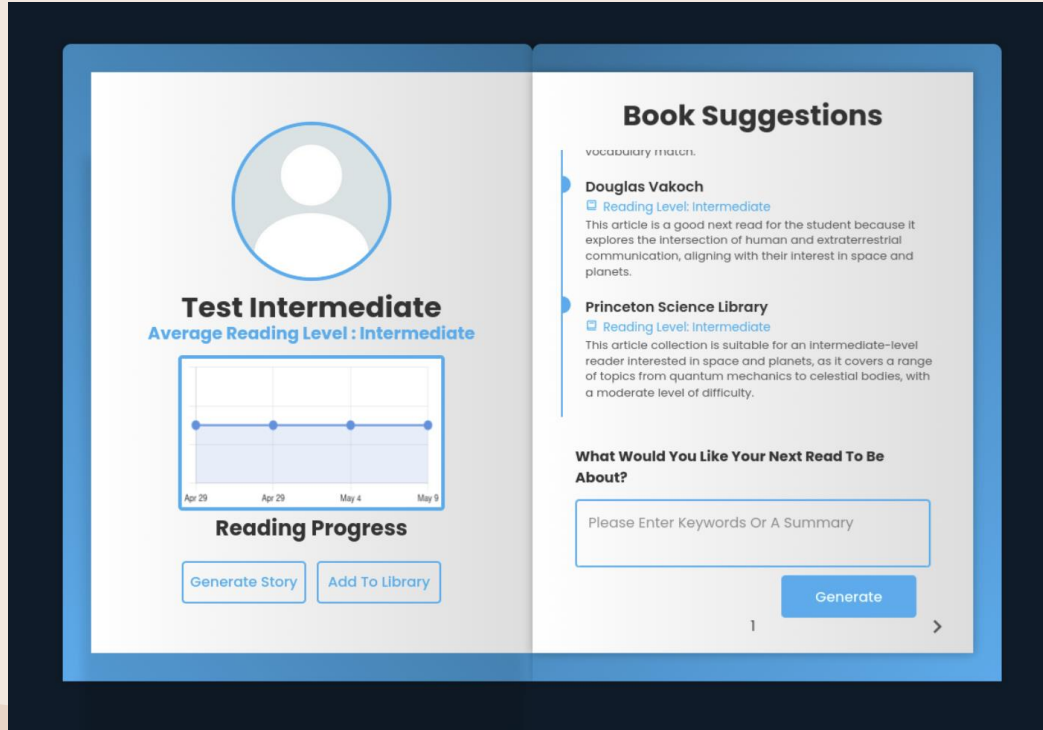
03 | Profile Page

User's profile page & suggestion page



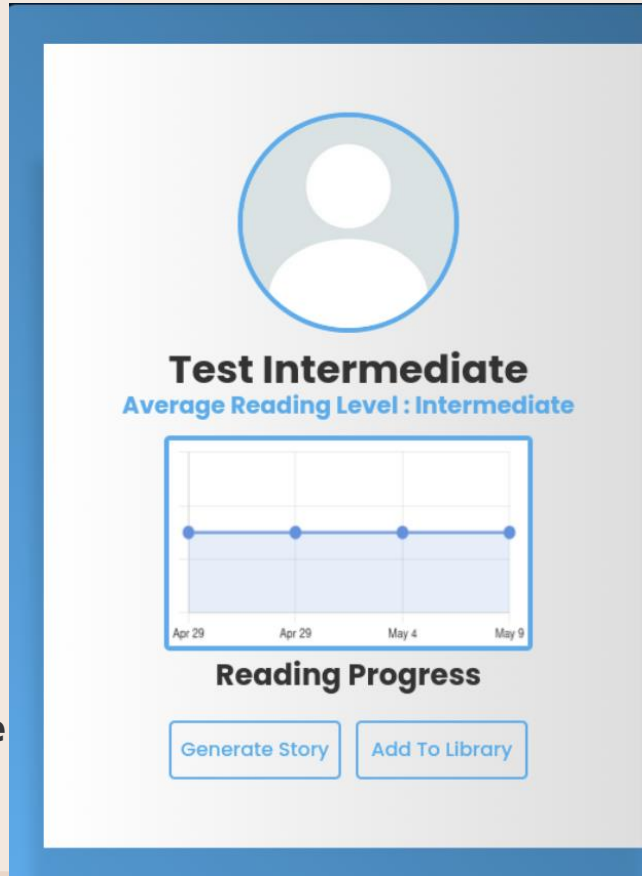
PROFILE PAGES

When the book opens, the first two pages greeting the user are the main profile page and the book suggestions page.



There's 3 different reading levels:

- **Beginner**
(grades K-4)
- **Intermediate**
(grades 5-8)
- **Advanced**
(grades 9+)



PROFILE PAGE

On the left, the user will see their **name** ("Test Intermediate") and their **average reading level**.

As the user adds more books to their library, their **Reading Progress graph** will update, showing if the book was advanced or basic based off their current reading level.

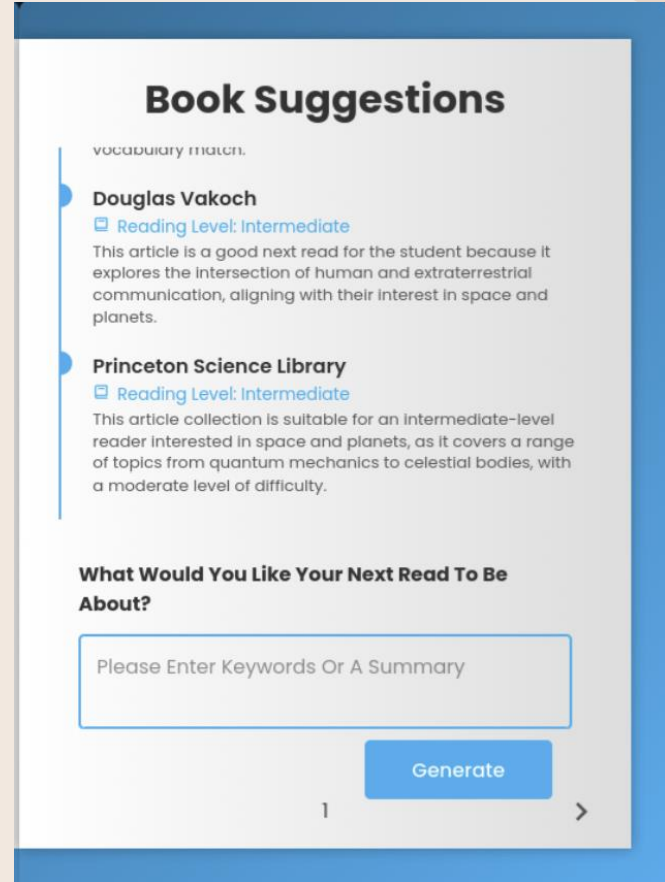
To quickly move around, the user can press the **generate story** or the **Add To Library** buttons that'll have them jump to the pages.

BOOK SUGGESTIONS

On the right, the user will see 3 book suggestions. The book suggestions are able to be read using a scroll wheel (which appears when you hover over it), allowing the user to see the **book's title**, **reading level**, and a **short description**.

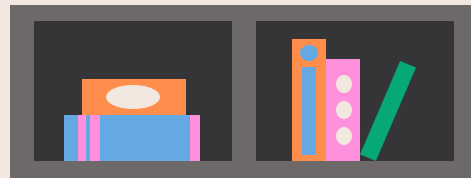
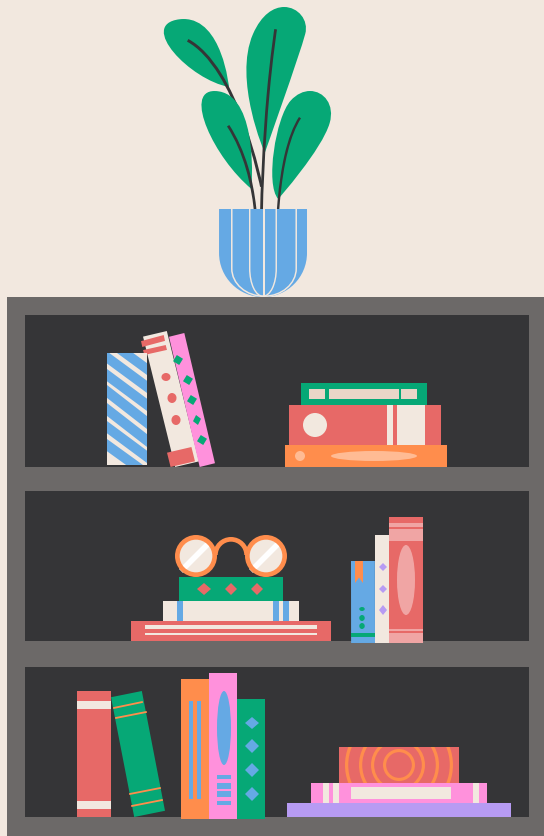
If the user wants a more personalized reading suggestion, they can enter keywords into the **textbox** and press **generate** in order to get new suggestions.

If the user wants to flip to the next page, they can press the **arrow** in the bottom right corner.



04

Personal Library .

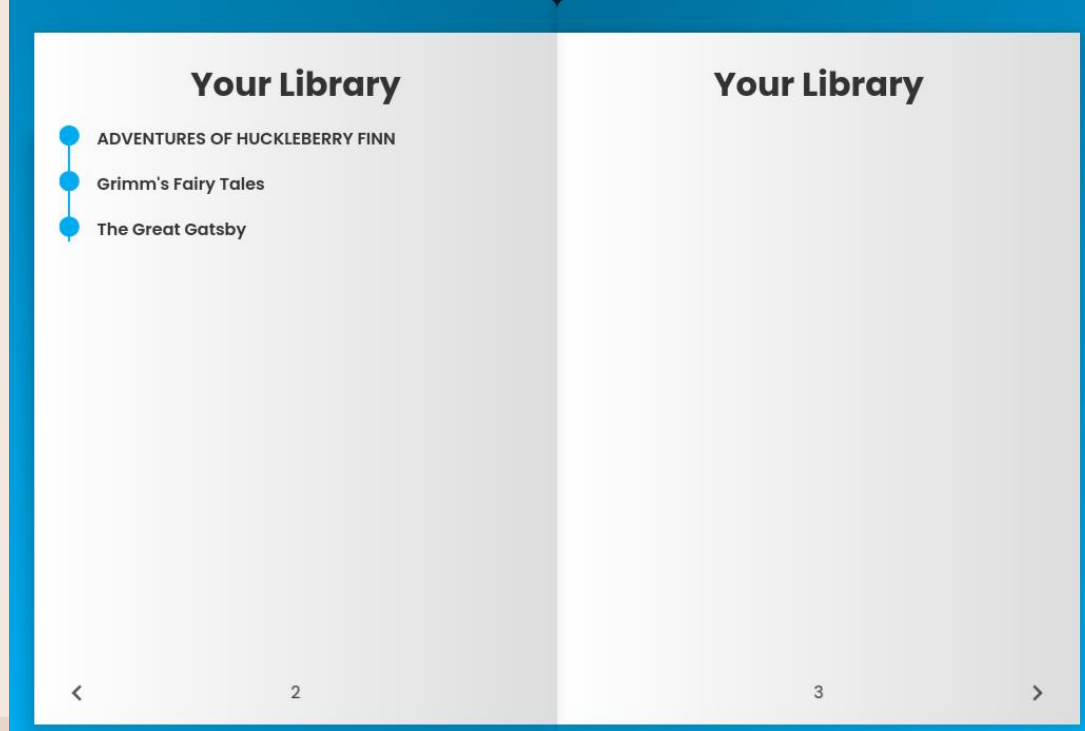


PERSONAL LIBRARY

When the user flips the page, the user can view their own personal library.



- Pressing the **arrow** in the left-hand corner will flip back to their profile. Meanwhile, pressing the **arrow** on the right-hand side will flip to the next pages.



Each time they add something to their library (pages 4 & 5), it's added to their "read books" in the database, which will be printed here.



05

ADD TO LIBRARY



ADD TO LIBRARY

There's two ways for the user to be able to add books into their personal library.

Add Your Own Book

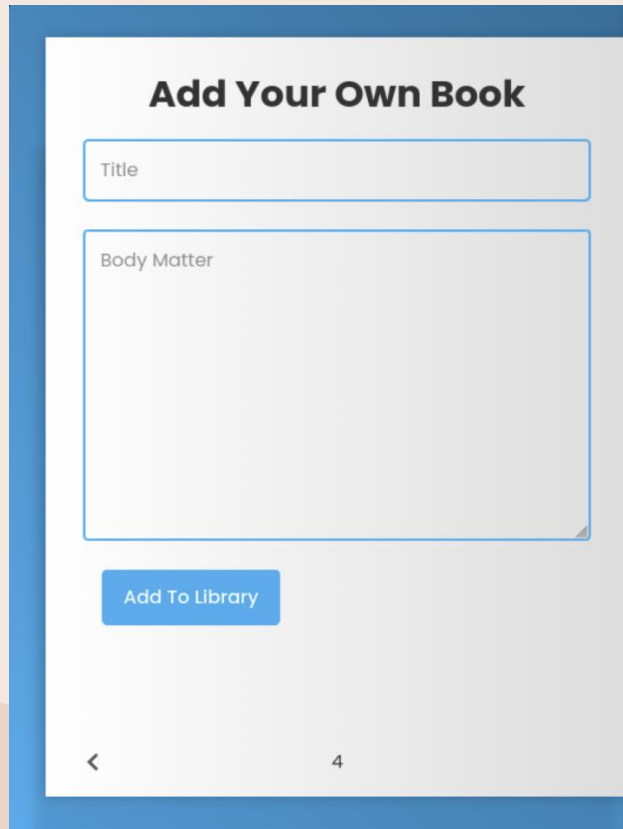
Body Matter

Add To Library

Search For Book

Please Input Your Book's Title That You'd Like To Search For:

Search



Add Your Own Book

Title

Body Matter

Add To Library

< 4

ADD YOUR OWN BOOK

On the left, the user can enter their own book (or a book's text script they might have found online).

To add the book to their library, the user will need to enter in the **title** and the book's **body matter** before pressing the **Add To Library** button. If they try to press it without entering anything, they'll receive an error message.

This book is then added to their personal library, and the program determines the reading level based off of its text.

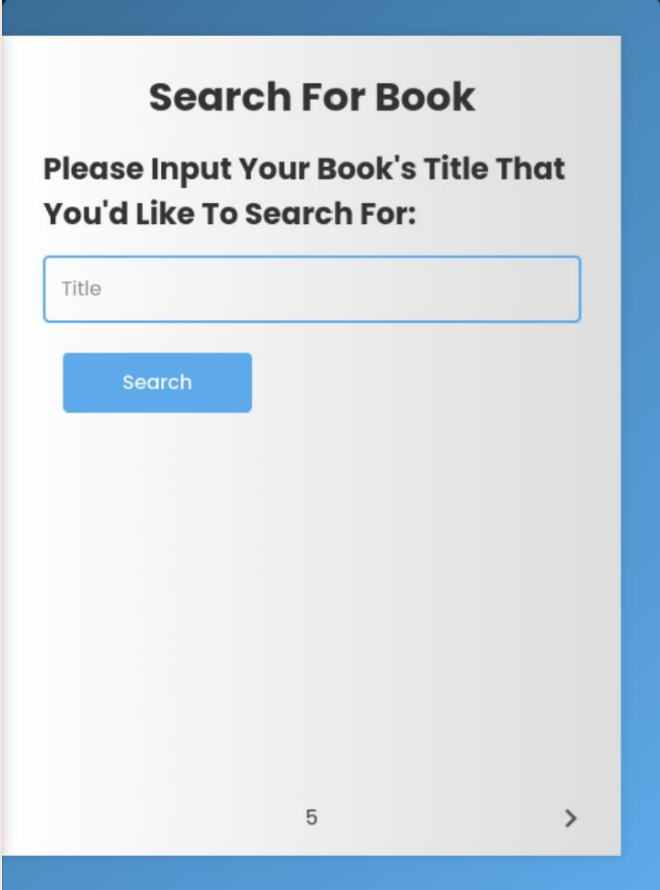
SEARCH FOR BOOK

On the right, the user will be able to search for a book that's already in the program's database based off the book's **title**.

This could be a book that is mentioned in one of the Wikipedia articles.

- As the program only has ~1 million scraped articles, the user may need to copy and paste the book & its text into the "Add Your Own Book" page, but it's still a nice option to search in case.

If there is a successful search, then the book is added to the user's library.



Search For Book

Please Input Your Book's Title That You'd Like To Search For:

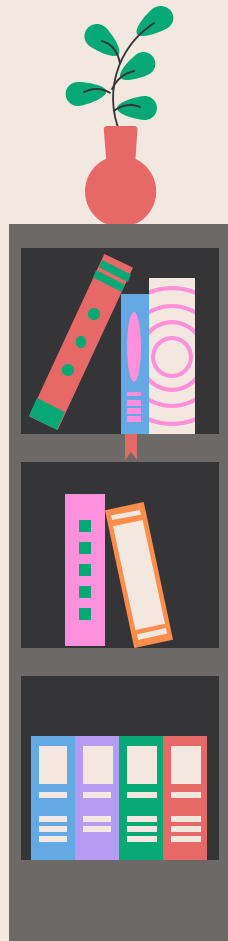
Search

5 >

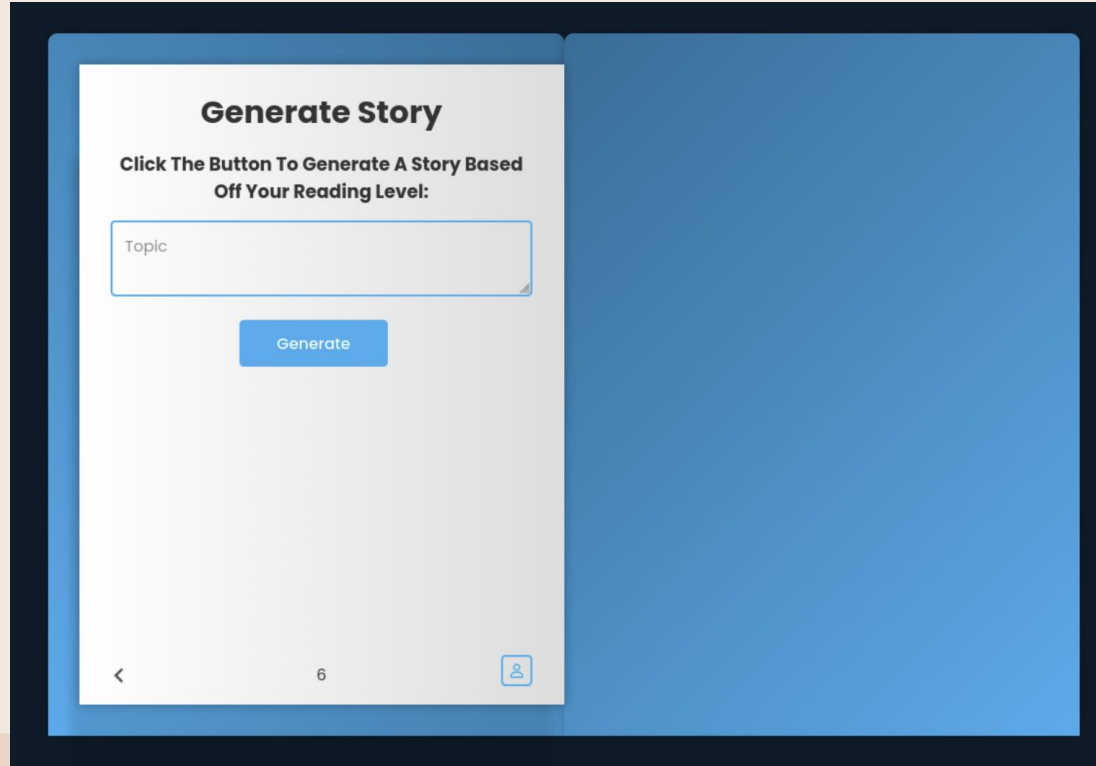
06

Generate Story

Generating story page based off the user's reading level



GENERATE STORY

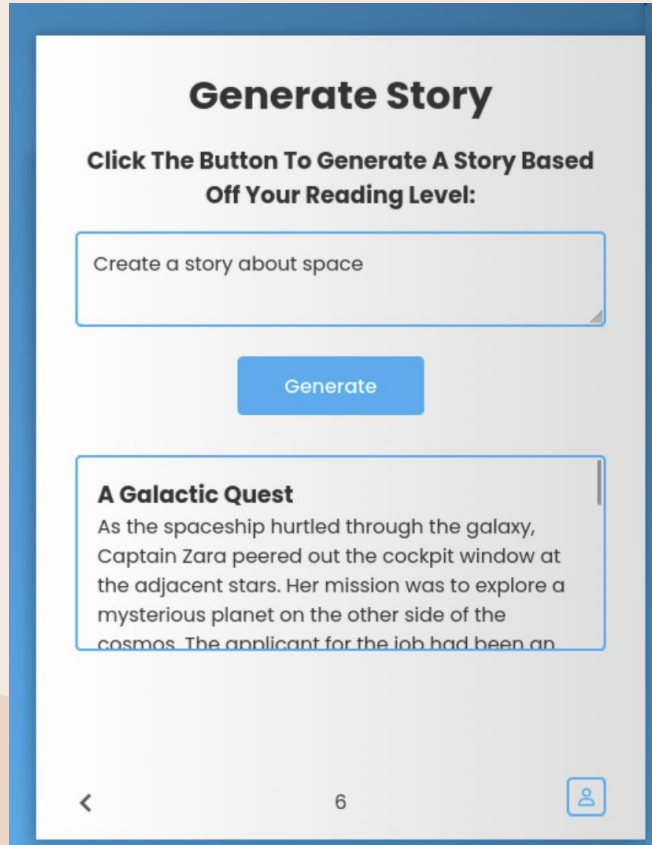


The screenshot shows a mobile application interface with a dark blue header and a light blue background. The main content area is white and contains the following elements:

- Generate Story**: A bold black title at the top of the white area.
- Click The Button To Generate A Story Based Off Your Reading Level:**: A line of black text below the title.
- Topic**: A text input field with a light blue border and a small blue icon in the bottom right corner.
- Generate**: A blue button with white text centered below the input field.
- Navigation Bar**: A dark blue bar at the bottom of the white area containing a back arrow, the number 6, and a user profile icon.

The right side of the screenshot is a solid blue rectangle, likely representing the generated story content.

The last page allows the user to generate a story based off keywords/topics and their reading level.



The screenshot shows a mobile application interface with a blue border. At the top, the title "Generate Story" is displayed in bold black text. Below it, a subtitle reads "Click The Button To Generate A Story Based Off Your Reading Level:". A text input box contains the prompt "Create a story about space". Below the input box is a blue button labeled "Generate". Underneath the button, a scrollable text box displays a generated story titled "A Galactic Quest". The story text reads: "As the spaceship hurtled through the galaxy, Captain Zara peered out the cockpit window at the adjacent stars. Her mission was to explore a mysterious planet on the other side of the cosmos. The applicant for the job had been an". At the bottom of the screen, there is a navigation bar with a left arrow, the number "6", and a person icon.

Generate Story

Click The Button To Generate A Story Based Off Your Reading Level:

Create a story about space

Generate

A Galactic Quest

As the spaceship hurtled through the galaxy, Captain Zara peered out the cockpit window at the adjacent stars. Her mission was to explore a mysterious planet on the other side of the cosmos. The applicant for the job had been an

< 6 

GENERATE STORY

If the user wants a personalized story to read, they can enter in the **text box** keywords or a prompt. When they hit the **generate** button, they'll need to wait about a minute for the software to develop a story. Once the software is done, a **box with a scroll wheel** will appear below.

This uses generative ai in order to create a personal written story. The short story will be based off of the user's known words (found through their other books in their library) and their average reading level.

In order to quickly jump back to their profile page, the user can click the **person icon** in the bottom right of the left page. If they want to flip back to the previous pages one-by-one, they can press the arrow in the bottom left.

○ THANKS!

DO YOU HAVE ANY QUESTIONS?

Github Link:

<https://github.com/KellyLFrear/MyLittleLibrary>

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